Shlok Thakkar

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Education

University of Illinois Urbana-Champaign — Champaign, IL
Bachelor of Science in Computer Science + Economics, Minor in Statistics | GPA: 3.97/4.00 — *Expected May 2028*

Technical Skills

Languages: Swift | C/C++ | Python | TypeScript | Go | SQL | Java | R
Frameworks: SwiftUI | Combine | AVKit | FastAPI | Flask | Next.js | LangChain
Databases: PostgreSQL | Neo4j | Supabase | MongoDB
Cloud & DevOps: AWS (Bedrock, S3, EC2) | Azure Blob | Docker | Git | CMake
Libraries: NumPy | Pandas | SciPy | Scikit-learn | lxml

Experience

Undergraduate Research Assistant — Apr. 2026 – Present
Parallel Programming Laboratory, University of Illinois Urbana-Champaign — *Champaign, IL*
- Replaced SUMMA with Cannon's algorithm and a circular-shift communication pattern, cutting per-step message startup overhead in distributed matrix multiplication across multi-node Charm++ clusters.
- Accelerated framework capabilities by extending Charm++'s distributed array abstraction with custom operators, relational and logical operations, and broadcasting support for high-performance parallel workloads.

Software Developer — Sept. 2025 – Present
Disruption Lab @ UIUC — *Champaign, IL*
- Delivered a Python-based XSD parser for a Fortune 500 financial services client's 44-jurisdiction tax schema pipeline using recursive lxml traversal and XLSX business rules ingestion.
- Expanded schema diff coverage from 7 to 27 states (+286%) through version-based XSD deduplication, and validated correctness with 43 unit tests across all fixture and mapping cases.
- Increased platform utility for 40+ users by building a REST API in Python (Flask) and Neo4j that exposed real-time project-resource dependency graphs, converting a read-only graph into a fully editable system.

Undergraduate Research Assistant — May 2025 – Dec. 2025
Department of Finance, University of Illinois Urbana-Champaign — *Champaign, IL*
- Enabled cross-database empirical analysis for Prof. Tatyana Deryugina's research by designing a custom NLP pipeline combining TF-IDF vectorization, Levenshtein distance, and fuzzy matching to link firms across 10M+ records.
- Reduced entity resolution runtime by 73% by introducing batched name normalization, deduplication, and text cleaning prior to fuzzy matching, processing the full corpus in under 4 hours.
- Achieved a 63% firm-level match rate across TED and ORBIS, directly unlocking the dataset needed for large-scale cross-database empirical research.

Software Development Intern — June 2025 – Aug. 2025
IQM Corporation — *Ahmedabad, India*
- Reduced P95 response latency 40% by architecting an async LLM pipeline on Amazon Bedrock with Pydantic-validated JSON extraction, serving 40+ enterprise users under 20+ concurrent requests.
- Shipped a FastAPI microservice on AWS EC2 processing 10,000+ candidate profiles via versioned REST endpoints with JWT authentication, rate limiting, and structured error handling.
- Built a CSV ingestion pipeline with AWS S3 integration persisting structured outputs for downstream ML candidate matching workflows across 10,000+ profiles.

Projects

FeeLens | *SwiftUI, Combine, AVKit, Supabase, PostgreSQL | Github* — Apr. 2026
- Built a real-time 1v1 iOS doomscrolling game at HackIllinois using SwiftUI, Combine, and Supabase with telemetry-driven competitive scoring.
- Implemented matchmaking and live leaderboard sync over Supabase PostgreSQL RPC functions with anonymous JWT auth and proactive token refresh.
- Designed a TTL-based feed cache with stale-while-revalidate, in-flight request deduplication, and adaptive score polling to minimize latency.

Limit Order Book | *C++20, CMake, Catch2 | Github* — Mar. 2026
- Engineered a thread-safe C++20 limit order book supporting 5 order types (GTC, FOK, FAK, GFD, Market) with FIFO price-time priority matching across concurrent producer and consumer threads.
- Achieved O(1) order cancellation by storing per-order iterators into price-level linked lists, eliminating linear scans and reducing worst-case cancel latency by orders of magnitude.
- Built an incremental LevelData cache reducing FOK pre-check complexity from O(N) to O(P), a ∼500x improvement for books with sparse price levels.

Awards

2nd Place — HERE Chicago Hackathon | *Geospatial ML model for real-time predictive mapping* — Apr. 2026
2nd Place — HackIllinois 2026 | *Scroll Royale, iOS doomscrolling game, SwiftUI + Supabase* — Feb. 2026